

TEZPUR UNIVERSITY

Autumn Semester End Examination, December, 2020

CO434/CS602: Image Processing

Full Marks: 40

Time: 2 hours

Instructions: *Maximum 30 minutes extra time is allowed to upload the scan copy of the answer sheets. Put your signature, roll number and page number in every answer sheet.*

1. Fill up the blanks in the following statements (answer any 5): 1×5=5
 - a) The number of shades of gray in a six-bit image is ____.
 - b) Order statistic filters are used for image ____.
 - c) Outwardly pointing corners are rounded by the ____ operation.
 - d) Ringing affect is not present in ____ lowpass filters
 - e) Sum of the weights in a smoothing filter is always ____.
 - f) The splitting technique used in image segmentation can be represented using a ____ tree.
2. Provide very brief to the point answers for the following questions (any five): 1×5=5
 - a) Present hexadecimal codes for two grey scale all system safe colours.
 - b) Present the filter masks for Sobel edge detector.
 - c) Find the complement of the colour represented by (200, 100, 0)
 - d) Present the equation for computing discrete Fourier transform of a two dimensional image $f(x, y)$.
 - e) What is the use of hit-or-miss transform?
 - f) State uses of the max filter.
3. Write very brief answers (any five): 2×5=10
 - a) How are isolated points recognized?
 - b) Simply describe a subtractive colour model.
 - c) How can you enhance the contrast of an image?
 - d) What are the different approaches for image segmentation?
 - e) Differentiate between filtering in spatial domain and frequency domain.
 - f) Present the convolution masks for line detection.
4.
 - a) What is an edge in an image? 1
 - b) On what mathematical operations are the basic approaches for edge detection based on? 1
 - c) What are different edge models. 1
 - d) Describe one method for edge detection. 2
5.
 - a) What is the basic mathematical operation that is used for image smoothing? 1
 - b) Describe methods for performing image smoothing in spatial as well as frequency domains. 4

6. Perform dilation and erosion on the region consisting of 1s given in the binary image shown on the left with the structuring element shown on the right. The shaded cell denotes the origin of the structuring element. 2+2=4

0	1	1	1	0	1
0	1	1	0	0	0
0	1	0	1	1	0

1	1
1	

7. Consider the image given below:

2	4	8	7
12	6	9	8
10	9	4	8

- a) Find the DC component of the Fourier transform of the image. 2
b) Show the binary image after segmenting with a threshold of 8. 2
c) Smooth the image by applying Median filter using the filter mask shown below. 2

